

IL-5 / IL-5R: 5 of 12 completed trials missed their primary endpoint

5 completed trials across 3 sponsors and 3 molecules ran to the end and missed their primary endpoint on the sponsor's own posted result; 7 met it. Only 2 of 34 closed trials were stopped early for efficacy or safety, which is why this class is read through its completed trials rather than a stop rate. Most completed trials post nothing readable, so the count is a floor.

5 of 12	3 / 3	5.9%	81.0%
completed trials with a readable primary result missed it · 7 met it	sponsors / molecules behind the misses	stopped early: 2 of 34 closed trials (95% CI 1.6%–19.1%)	of 42 trials have closed · 8 still open or unresolved

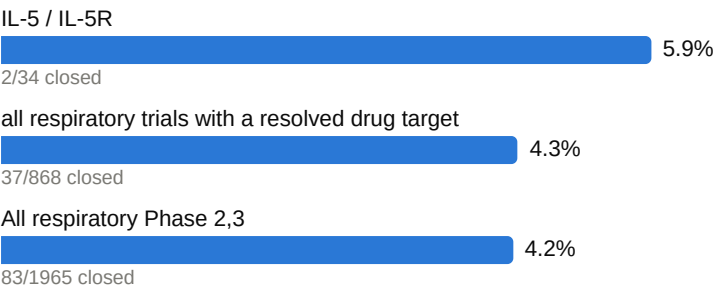
5 of 12

Completed and missed the primary endpoint. Of 30 completed trials in this segment, 12 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 5 missed and 7 met. Counted separately and never in the rate above; a floor, not a rate. 3 of the 28 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

The completed trials that missed

Trial	Ph	Sponsor	Experimental drugs	Started	What the posted result says
NCT04075331	3	University of Leicester	MEPOLIZUMAB	2020-09	Time From Randomisation to Next Hospital Readmission or Death (All Cause) — p 0.811, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Hazard Ratio (HR) 0.96 (95% CI 0.7 to 1.32). Method: Log Rank.
NCT02869438	3	AstraZeneca	BENRALIZUMAB	2016-11	Change From Baseline (Visit 4) to Day 28 (Visit 8), Day 56 (Visit 9), and Day 84 (Visit 10) in Pre-BD FEV1 — p 0.7747, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Mean Difference (Final Values) 0.013 (95% CI -0.077 to 0.104). Method: Mixed Models Analysis.
NCT02821416	3	AstraZeneca	BENRALIZUMAB	2016-10	Maximal Percentage Decrease in Forced Expiratory Volume in 1 Second 3-7 Hours Post Allergen Challenge — p 0.363, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Median Difference (Final Values) 2.535 (95% CI -3.045 to 8.116). Method: Mixed Models Analysis.
NCT02501629	3	Teva Branded Pharmaceutical Products R&D, Inc.	RESLIZUMAB	2015-09	Number of Participants With Reduction In Daily Oral Corticosteroid (OCS) Dose During Weeks 20-24 As Compared to the Optimized Dose At Baseline — p 0.468, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Odds Ratio (OR) 1.23 (95% CI 0.702 to 2.157). Method: proportional odds model.
NCT02452190	3	Teva Branded Pharmaceutical Products R&D, Inc.	RESLIZUMAB	2015-09	Number of Clinical Asthma Exacerbations (CAEs) During 52 Weeks of Treatment — p 0.194 against the sponsor's stated threshold of 0.05: not significant. CAE rate ratio (reslizumab vs placebo) 0.79 (95% CI 0.562 to 1.124). Method: Chi-squared.

How this compares



What is in this class: 5 distinct experimental drugs across 42 trials (antibody 42, small molecule 1). Most tested: BENRALIZUMAB (20), MEPOLIZUMAB (10), RESLIZUMAB (7), DEPEMOKIMAB (5), MOMETASONE FUROATE (1). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 2 stops came from 2 sponsor-asset programmes across 2 sponsors; the largest (Teva Branded Pharmaceutical Products R&D, Inc.) contributed 1; removing that programme's trials from both sides leaves 3.2%. Screened alongside 13 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. One further closed trial here was terminated with no cause recorded in the registry; if it were biological the rate would be 8.8%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04157335	3	AstraZeneca AZN	BENRALIZUMAB	2024-08	Efficacy	The study has a double-blind period and an open-label period. The study did not meet its primary endpoints after the primary LSLV and the decision was taken to terminate the study early.
NCT03052725	3	Teva Branded Pharmaceutical Products R&D, Inc.	RESLIZUMAB	2018-02	Efficacy	Parent Studies didn't meet their primary endpoint so study was terminated.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 respiratory trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 42% of industry respiratory trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	1	19	5.3%	0.9%–24.6%
2018–2020	1	8	12.5%	2.2%–47.1%
2021–2024	0	7	0.0%	0.0%–35.4%

Where the stops sit

Efficacy / futility: 2.
Sponsors with most stops: AstraZeneca (1), Teva Branded Pharmaceutical Products R&D, Inc. (1).

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: clinicaltrialfailures.com/asset-check